

US GDP Analysis

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Initial Data Processing

Importing Libraries

```
library(ggplot2)
library(dplyr)
library(tidyverse)
library(forecast)
library(strucchange)
library(corrplot)
library(psych)
library(moments)
library(e1071)
library(tseries)
library(car)
library(caret)
library(glmnet)
library(reshape2)
```

Data Overview (Schema)

- Year (Integer): Year data was recorded
- GDP_PPP (Decimal, 1): Value of all purchased final goods produced within different countries, that can be adjusted based on inflation rates
- GDP_PPP_PerCapita (Decimal, 1): Comprehensive GDP_PPP divided by population and adjusted for cost of living and inflation
- GDP_Nominal (Decimal, 1): Value of all purchased final goods, with no adjustment based on inflation
- GDP_Nominal_PerCapita (Decimal, 1): Comprehensive GDP_Nominal divided by population and adjusted for cost of living and inflation
- GDP_Growth (Decimal, 3): Growth Rate of GDP based on previous year
- Inflation (Decimal, 3): Increase rate of general prices of goods and services
- Unemployment (Decimal, 3): Percentage of individuals in US seeking for work
- Inflation_Change (String): Indicator of Increase or Decrease of Inflation based on current and prior inflation rate

Data Validation using SQL

```
# No Queries Actually Evaluated here, but done separately on a SQLite Server
```

```
SELECT Year
FROM Economy_of_US
WHERE GDP_Growth > 0.05; #Selects Years with GDP > 5%
```

```
SELECT
(
  SUM((GDP_PPP - avg_x) * (GDP_Nominal - avg_y)) -- Numerator: Sum of products of deviations
) / NULLIF(
  SQRT(
    SUM((GDP_PPP - avg_x) * (GDP_PPP - avg_x)) * SUM((GDP_Nominal - avg_y) * (GDP_Nominal - avg_y))
  )
)
```

```

    ),
    0
) AS correlation_coefficient
FROM
(
    SELECT
        t.*,
        AVG(GDP_PPP) OVER () AS avg_x,
        AVG(GDP_Nominal) OVER () AS avg_y
    FROM
        Economy_of_US t
) AS subquery_with_averages; #Calculates Correlation between GDP PPP and Nominal Metrics using the st

SELECT GDP_Nominal, Year, AVG(GDP_Nominal) OVER (
ORDER by Year
rows BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg
FROM Economy_of_US; -- 3 Year Moving Average for Nominal GDP

SELECT GDP_Nominal, Year, AVG(GDP_Nominal) OVER (
ORDER by Year
rows BETWEEN 4 PRECEDING AND CURRENT ROW)
AS moving_avg
FROM Economy_of_US; # 5 Year Moving Average for Nominal GDP

SELECT GDP_PPP, Year, AVG(GDP_PPP) OVER (
ORDER by Year
rows BETWEEN 2 PRECEDING AND CURRENT ROW)
AS moving_avg
FROM Economy_of_US; # 3 Year Moving Average for GDP PPP

SELECT GDP_PPP, Year, AVG(GDP_PPP) OVER (
ORDER by Year
rows BETWEEN 4 PRECEDING AND CURRENT ROW)
AS moving_avg
FROM Economy_of_US; # 5 Year Moving Average for GDP PPP

SELECT Year, avg_growth
FROM(SELECT Year, GDP_Growth, AVG(GDP_Growth) over (
ORDER BY Year
ROWS BETWEEN 9 PRECEDING and CURRENT ROW
) AS avg_growth -- Calculates the average growth rate between current and previous 9 years
FROM Economy_of_US)
order by avg_growth DESC
LIMIT 1; #Finds the decade with the ending year with the highest average growth rate

WITH mean AS (
    SELECT AVG(GDP_Nominal) AS avg_value
    FROM Economy_of_US
),

```

```

stddev AS (
  SELECT
    SQRT(
      SUM((t.GDP_Nominal - m.avg_value) *
        (t.GDP_Nominal - m.avg_value)
      )
      / (COUNT(t.GDP_Nominal) - 1)
    ) AS stddev_value
  FROM Economy_of_US t
  CROSS JOIN mean m
),
anomalies AS (
  SELECT
    t1.*,
    (t1.GDP_Nominal - m.avg_value) / s.stddev_value AS z_score
  FROM Economy_of_US t1
  CROSS JOIN mean m
  CROSS JOIN stddev s # Selects Anomaly values, based on how far they are from the average
)
SELECT *
FROM anomalies
WHERE ABS(z_score) > 2;

```

Data Validation

Importing and Cleaning Data

When Reading the Data, I recognized that the GDP Nominal and GDP_PPP Metrics were identical. To resolve this issue, I removed 2 of the duplicate fields and renamed the other 2 fields as “GDP” and “GDP_Per_Capita”

```

df <- read.csv("Economy_of_US.csv")
df <- df[, c(1, 2, 3, 6, 7, 8, 9)]
colnames(df)[2] <- "GDP"
colnames(df)[3] <- "GDP_PerCapita"

```

Descriptive Statistics

```

gdp_df <- df[, 2:3]
quant_vars <- df[, 2:6]

summary(quant_vars)

```

##	GDP	GDP_PerCapita	GDP_Growth	Inflation
## Min.	: 2857	Min. :12553	Min. :-0.03400	Min. :-0.00300
## 1st Qu.:	6430	1st Qu.:25120	1st Qu.: 0.01700	1st Qu.: 0.02000
## Median :	11837	Median :40524	Median : 0.02600	Median : 0.02800
## Mean :	13182	Mean :43192	Mean : 0.02446	Mean : 0.03221
## 3rd Qu.:	18328	3rd Qu.:57007	3rd Qu.: 0.03725	3rd Qu.: 0.03525

```
## Max.      :30282   Max.      :89546   Max.      : 0.07200   Max.      : 0.13500
## Unemployment
## Min.      :0.0370
## 1st Qu.:0.0485
## Median :0.0555
## Mean     :0.0605
## 3rd Qu.:0.0720
## Max.     :0.0970
```

```
# Returns a named vector of variances
sapply(quant_vars, var, na.rm = TRUE)
```

```
##           GDP GDP_PerCapita   GDP_Growth   Inflation Unemployment
## 6.111153e+07 4.578268e+08 3.845940e-04 5.603812e-04 2.686809e-04
```

```
#Analyze Skewness of Quantitative Variables
skew(quant_vars)
```

```
## [1] 0.5414018 0.4644225 -0.6995630 2.3386895 0.7057662
```

```
#Kurtosis
apply(quant_vars, 2, kurtosis, na.rm = TRUE)
```

```
##           GDP GDP_PerCapita   GDP_Growth   Inflation Unemployment
## -0.8180309  -0.8198000    1.2801250    6.8463908   -0.4494227
```

For Skewness, the metrics indicate the following:

- Highly skewed if the absolute value is greater than 1
- Moderately skewed if the absolute value is between 0.5 and 1
- Symmetric if the absolute value is less or equal to 0.5

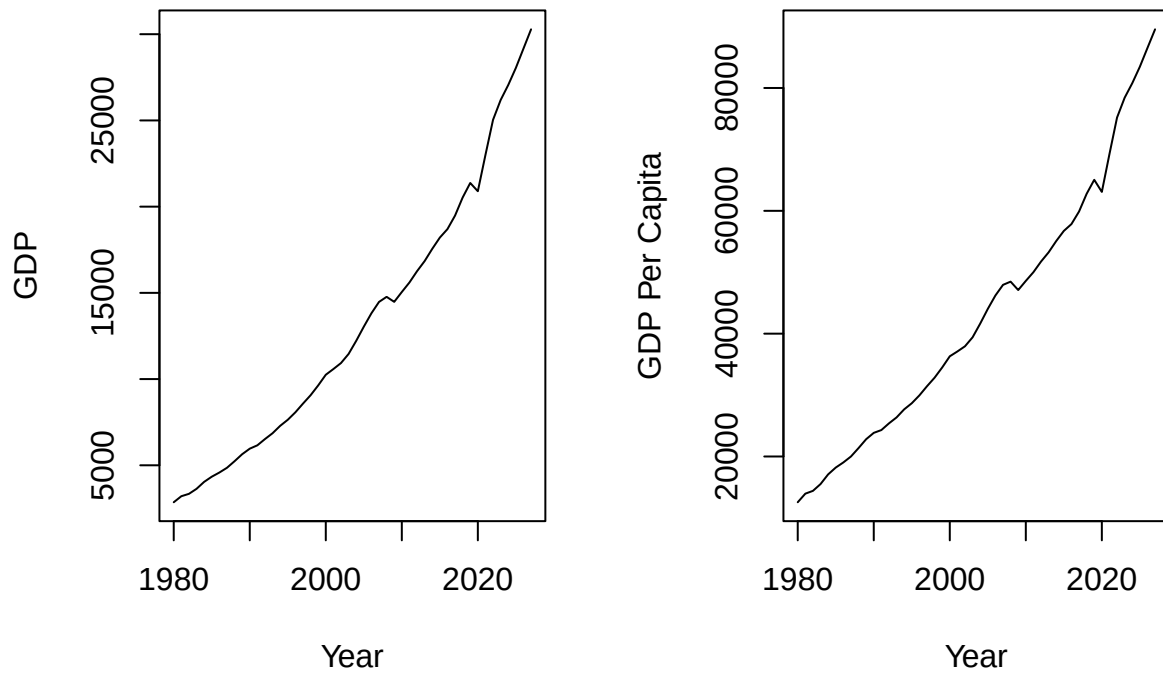
From the results, the each metric is distributed as follows:

- Unemployment is skewed
- GDP and GDP Per Capita is moderately skewed
- GDP Growth is Highly Skewed

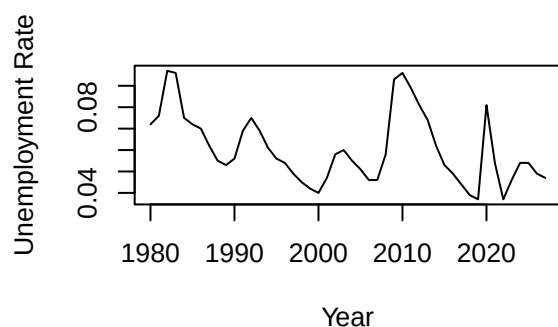
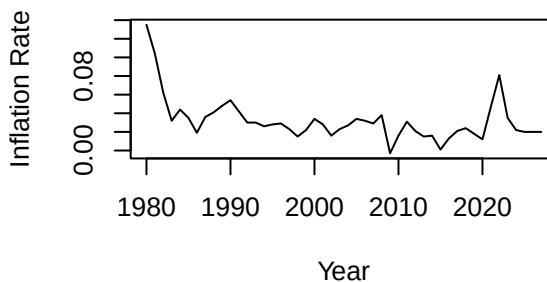
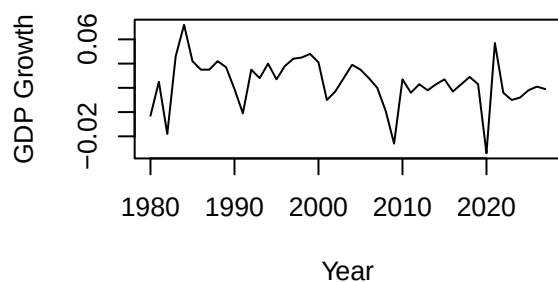
Visualizing Metrics Year After Year

```
par(mfrow = c(1, 2))

plot(df$Year, df$GDP, type = "l", xlab="Year", ylab = "GDP")
plot(df$Year, df$GDP_PerCapita, type = "l", , xlab="Year", ylab = "GDP Per Capita")
```



```
par(mfrow = c(2, 2))
plot(df$Year, df$GDP_Growth, type = "l", xlab="Year", ylab = "GDP Growth")
plot(df$Year, df$Inflation, type = "l", xlab="Year", ylab = "Inflation Rate")
plot(df$Year, df$Unemployment, type = "l", xlab="Year", ylab = "Unemployment Rate")
```



ANOVA

First, I added a column to label years based on decades. Here, I classified the years are based on the shared “Tens Digit”, and assigned a decade as the starting year it is part of.

```
df$decade <- as.character(floor(df$Year/10)*10)
```

```
anova_model <- aov(GDP ~ decade, data = df)
summary(anova_model)
```

```
##           Df    Sum Sq  Mean Sq F value Pr(>F)
## decade      4 2.714e+09 678619883    185 <2e-16 ***
## Residuals   43 1.578e+08  3668889
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The most important result to determine the Hypothesis is the P Value. Here, there is a Null Hypothesis, which states there is a difference between at least 2 decade GDP averages. If there is a 5% allowed chance of error in the data, then because the p-value is significantly less than 0.05, the null hypothesis is rejected and ANOVA concludes that the average GDP across all decades is not equal

Correlation Analysis

```
corrplot(cor(quant_vars), type = "upper", order = "hclust",  
         tl.col = "black", tl.srt = 45)
```



Time Series Analysis

Decomposing to Time Series Components

```
# Convert to time series  
gdp_ts <- ts(df$GDP, start = min(df$Year))  
  
print(gdp_ts)
```

```
## Time Series:  
## Start = 1980  
## End = 2027  
## Frequency = 1  
## [1] 2857.3 3207.0 3343.8 3634.0 4037.7 4339.0 4579.6 4855.3 5236.4  
## [10] 5641.6 5963.1 6158.1 6520.3 6858.6 7287.3 7639.8 8073.1 8577.6  
## [19] 9062.8 9631.2 10251.0 10581.9 10929.1 11456.5 12217.2 13039.2 13815.6
```



```
## [28] 14474.3 14769.9 14478.1 15049.0 15599.7 16254.0 16843.2 17550.7 18206.0
## [37] 18695.1 19479.6 20527.2 21372.6 20893.8 22996.1 25035.2 26185.2 27057.2
## [46] 28045.3 29165.5 30281.5
```

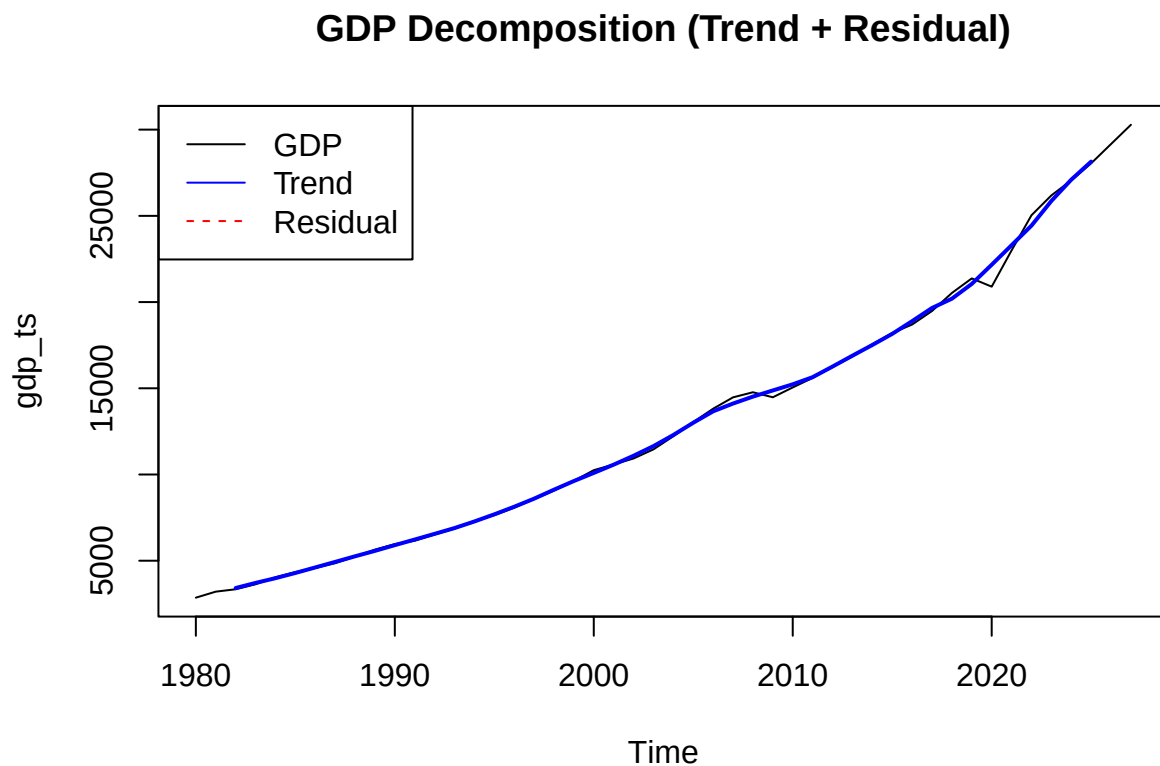
```
trend <- stats::filter(gdp_ts, rep(1/5, 5), sides = 2)

residual <- gdp_ts - trend

# Plotting the Trend and Residuals together, followed by a legend

plot(gdp_ts, main = "GDP Decomposition (Trend + Residual)", col = "black")
lines(trend, col = "blue", lwd = 2)
lines(residual, col = "red", lty = 2)

legend("topleft",
      legend = c("GDP", "Trend", "Residual"),
      col = c("black", "blue", "red"),
      lty = c(1,1,2))
```

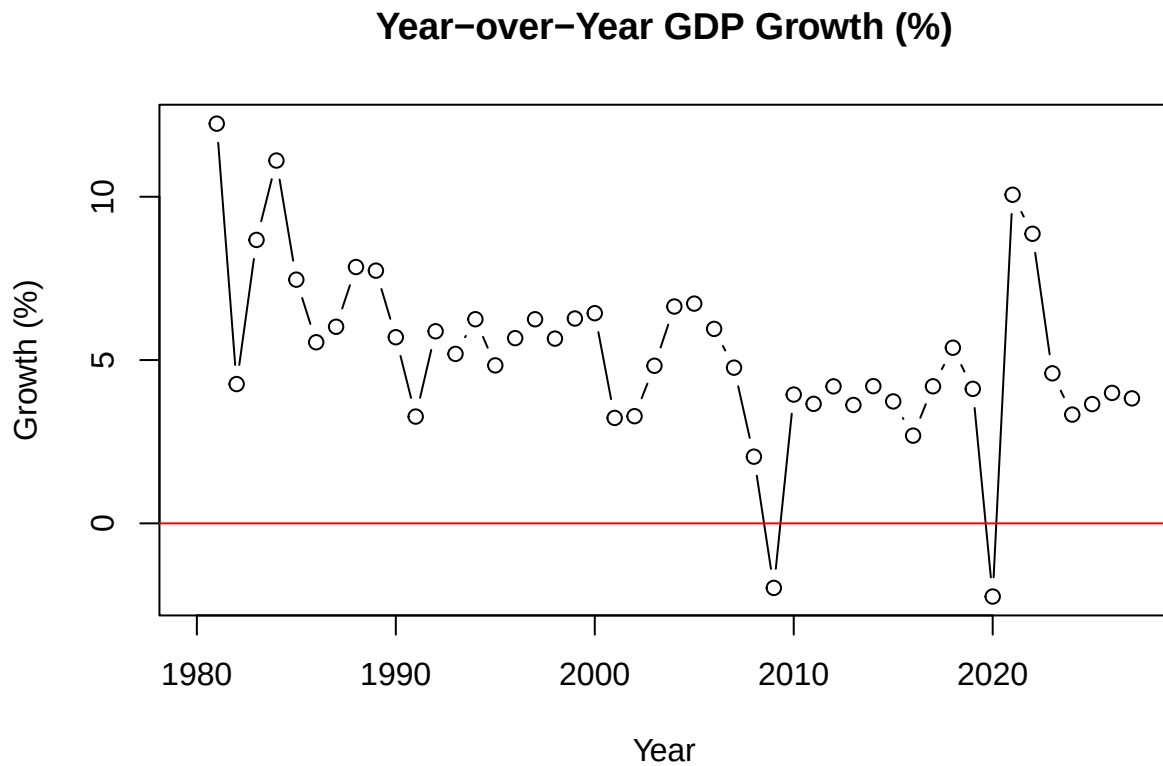


```
## Plotting and Observing Nominal GDP Year By Year
```

```
# Year vector
years <- 1980:2027

# YoY growth
yoy_growth <- c(NA, diff(as.numeric(gdp_ts)) / head(as.numeric(gdp_ts), -1) * 100)
```

```
# Plotting Points of Growth Rates Year after Year
plot(years, yoy_growth, type = "b",
     main = "Year-over-Year GDP Growth (%)",
     xlab = "Year", ylab = "Growth (%)")
abline(h = 0, col = "red")
```



Predictive Modelling

Transforming and Grouping Training and Testing Data

```
df <- df %>%
  mutate(
    Year_Scaled = Year - min(Year),      # normalize year
    Year_Sq = Year_Scaled^2,             # polynomial term
    Lag1 = lag(GDP, 1),                  # lag feature
    Growth = (GDP / Lag1 - 1) * 100
  ) %>%
  drop_na()

set.seed(123)

train_index <- createDataPartition(df$GDP, p = 0.8, list = FALSE)
```

```
train <- df[train_index, ]
test <- df[-train_index, ]
```

Linear, Polynomial, Ridge and Lasso Modelling

```
lm_model <- lm(GDP ~ Year_Scaled + Lag1 + Growth, data = train) #Initial Linear Regression Model
summary(lm_model)
```

```
##
## Call:
## lm(formula = GDP ~ Year_Scaled + Lag1 + Growth, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -474.91  -33.96   -4.99   46.25  398.40
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -853.13153   111.99291   -7.618 6.21e-09 ***
## Year_Scaled   27.35601    10.03659    2.726 0.00995 **
## Lag1          1.00349     0.01768   56.773 < 2e-16 ***
## Growth       142.62672    13.79385   10.340 3.51e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 160.4 on 35 degrees of freedom
## Multiple R-squared:  0.9996, Adjusted R-squared:  0.9996
## F-statistic: 2.939e+04 on 3 and 35 DF, p-value: < 2.2e-16
```

```
# Predictions
lm_pred <- predict(lm_model, newdata = test)

poly_model <- lm(GDP ~ Year_Scaled + Year_Sq, data = train)
poly_pred <- predict(poly_model, newdata = test)

# Prepare matrix format
x_train <- model.matrix(GDP ~ Year_Scaled + Year_Sq + Lag1 + Growth, train)[,-1]
y_train <- train$GDP

x_test <- model.matrix(GDP ~ Year_Scaled + Year_Sq + Lag1 + Growth, test)[,-1]
y_test <- test$GDP

# Ridge (alpha = 0)
ridge_model <- cv.glmnet(x_train, y_train, alpha = 0)
ridge_pred <- predict(ridge_model, s = "lambda.min", newx = x_test)

# Lasso (alpha = 1)
lasso_model <- cv.glmnet(x_train, y_train, alpha = 1)
lasso_pred <- predict(lasso_model, s = "lambda.min", newx = x_test)
```

```
rmse <- function(actual, pred) {
  sqrt(mean((actual - pred)^2))
}

results <- data.frame(
  Model = c("Linear", "Polynomial", "Ridge", "Lasso"),
  RMSE = c(
    rmse(y_test, lm_pred),
    rmse(y_test, poly_pred),
    rmse(y_test, ridge_pred),
    rmse(y_test, lasso_pred)
  )
)

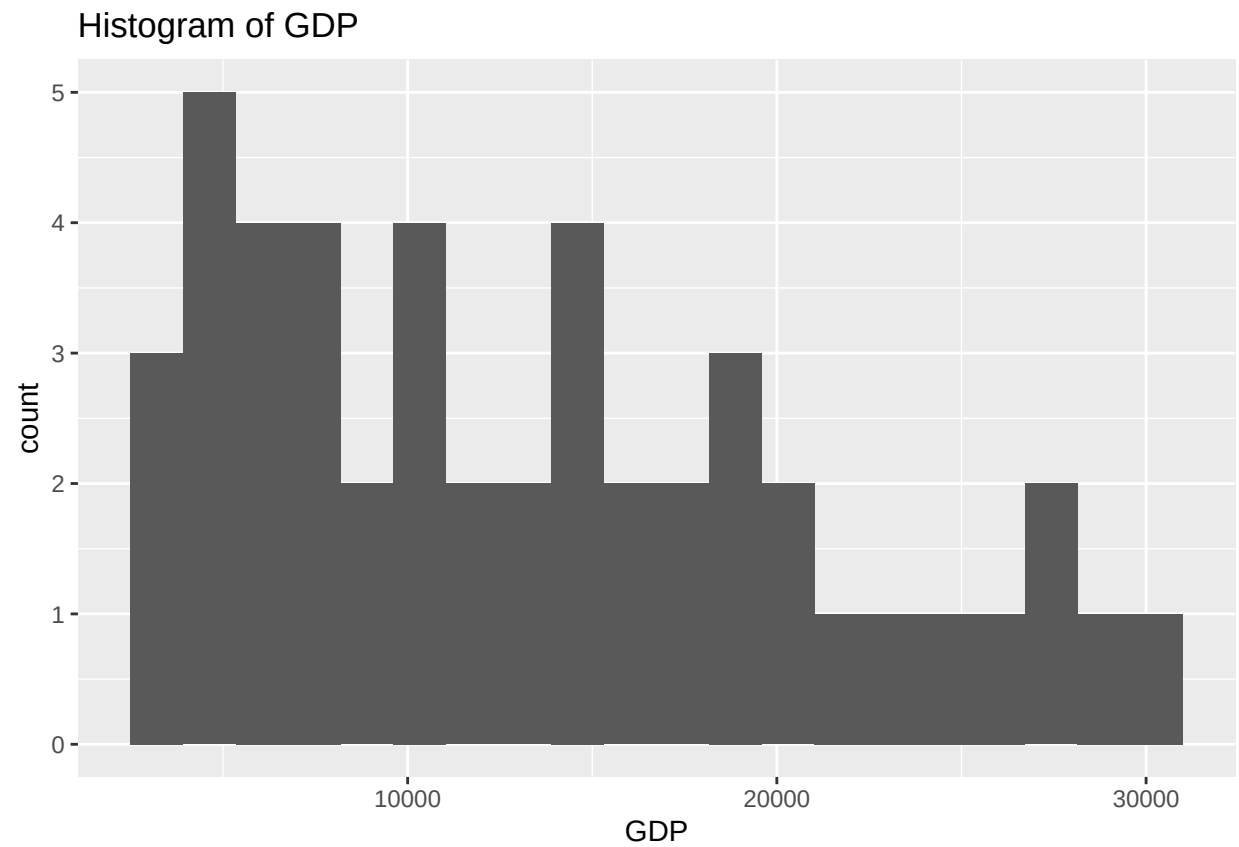
print(results)
```

```
##      Model      RMSE
## 1   Linear 238.7048
## 2 Polynomial 516.7115
## 3    Ridge 433.0670
## 4    Lasso 421.1631
```

Data Visualizations

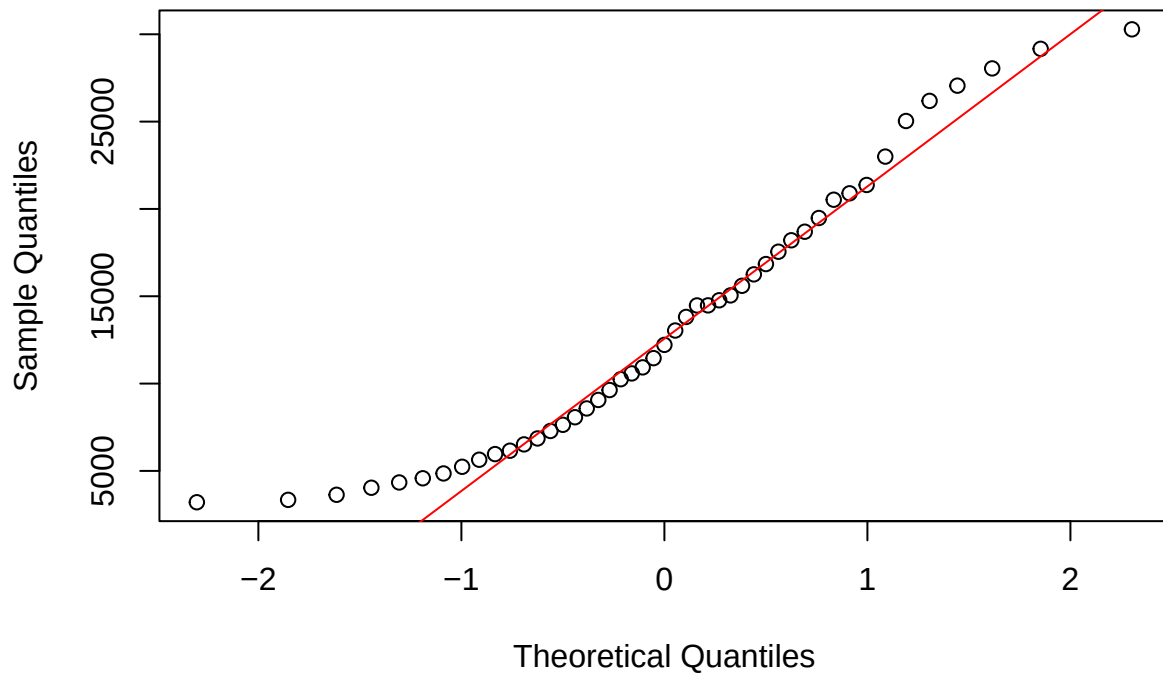
GDP Distribution

```
# Histogram
ggplot(df, aes(x = GDP)) +
  geom_histogram(bins = 20) +
  labs(title = "Histogram of GDP")
```



```
# QQ Plot  
qqnorm(df$GDP)  
qqline(df$GDP, col = "red")
```

Normal Q-Q Plot



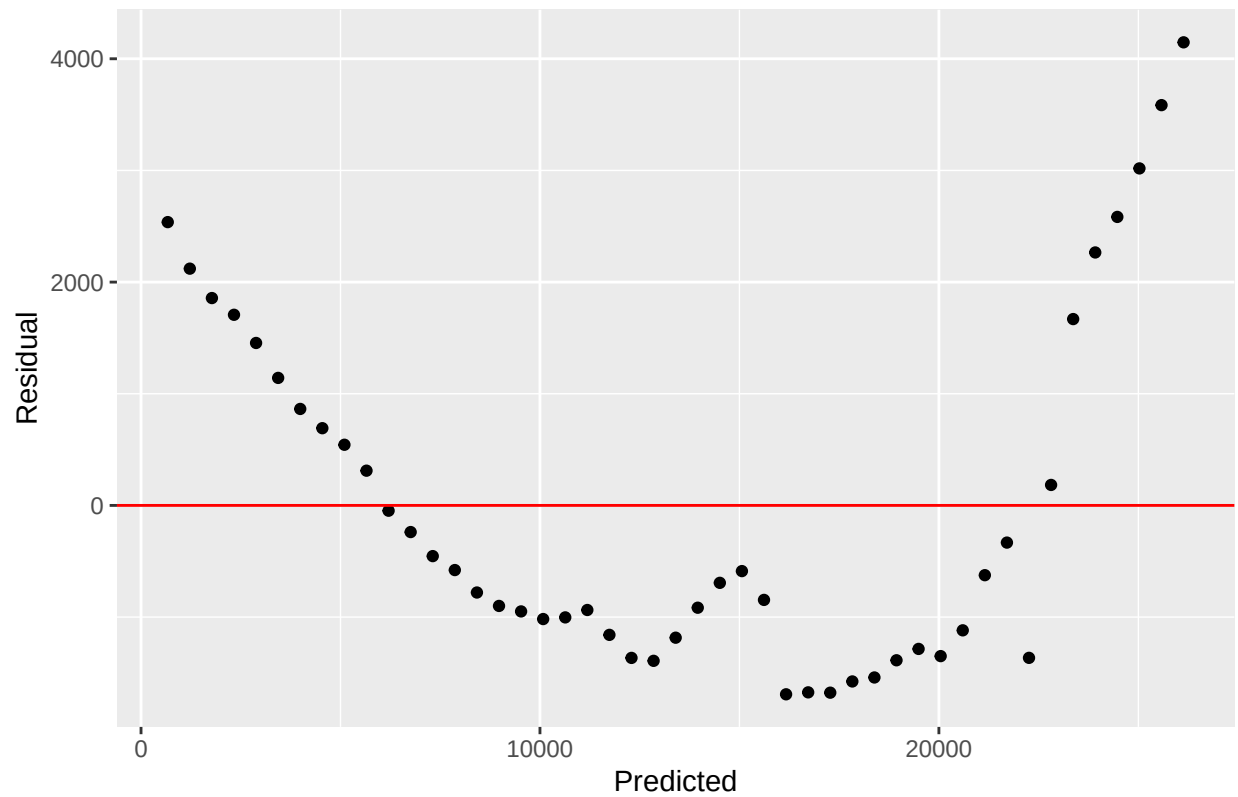
The QQ plot is intended to assess if the given time series data for GDP is normal. As most of the points form a line, it indicates the GDP has a normal distribution. This is expected since the histogram also forms a bell-shaped curve.

Regression Diagnostics

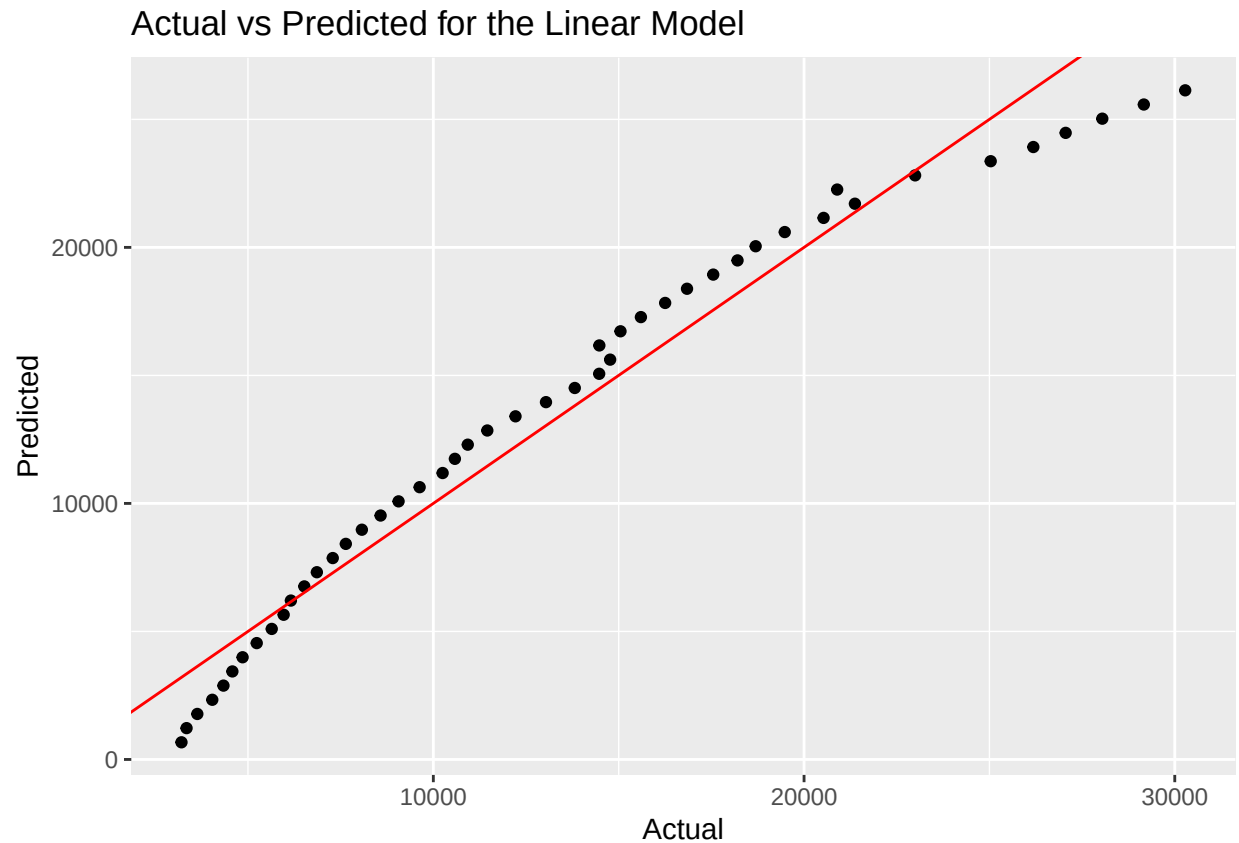
```
model <- lm(GDP ~ Year, data = df)
df$pred <- predict(model)
df$resid <- residuals(model)

ggplot(df, aes(x = pred, y = resid)) +
  geom_point() +
  geom_hline(yintercept = 0, color = "red") +
  labs(title = "Residuals vs Fitted for the Linear Model", x = "Predicted", y = "Residual")
```

Residuals vs Fitted for the Linear Model



```
ggplot(df, aes(x = GDP, y = pred)) +  
  geom_point() +  
  geom_abline(slope = 1, intercept = 0, color = "red") +  
  labs(title = "Actual vs Predicted for the Linear Model", x = "Actual", y = "Predicted")
```



Forecast Visualization

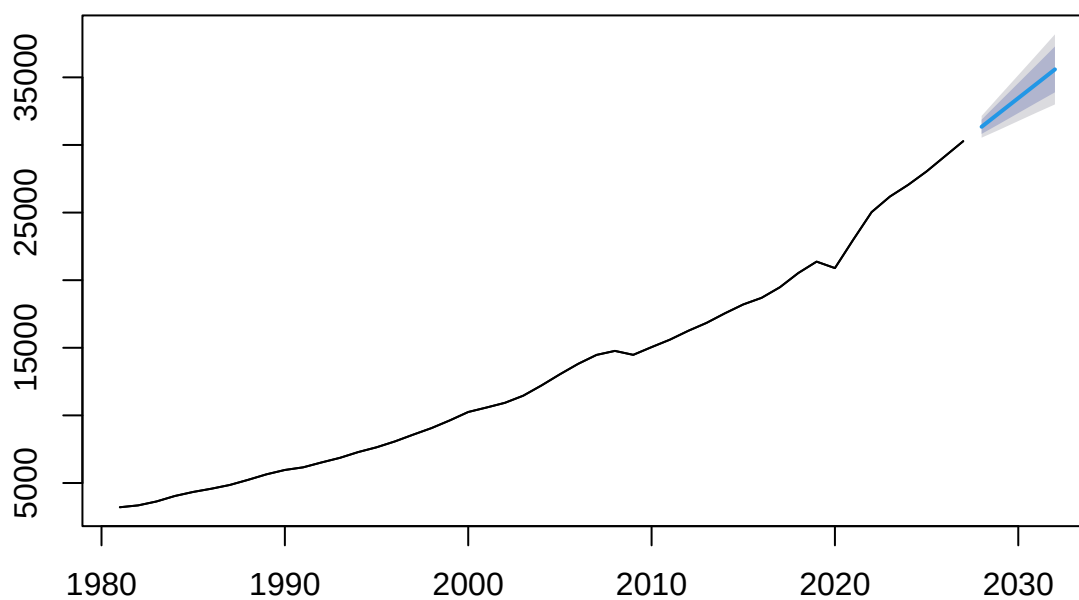
```
gdp_ts <- ts(df$GDP, start = min(df$Year), frequency = 1)

model_arima <- auto.arima(gdp_ts)

forecast_vals <- forecast(model_arima, h = 5)

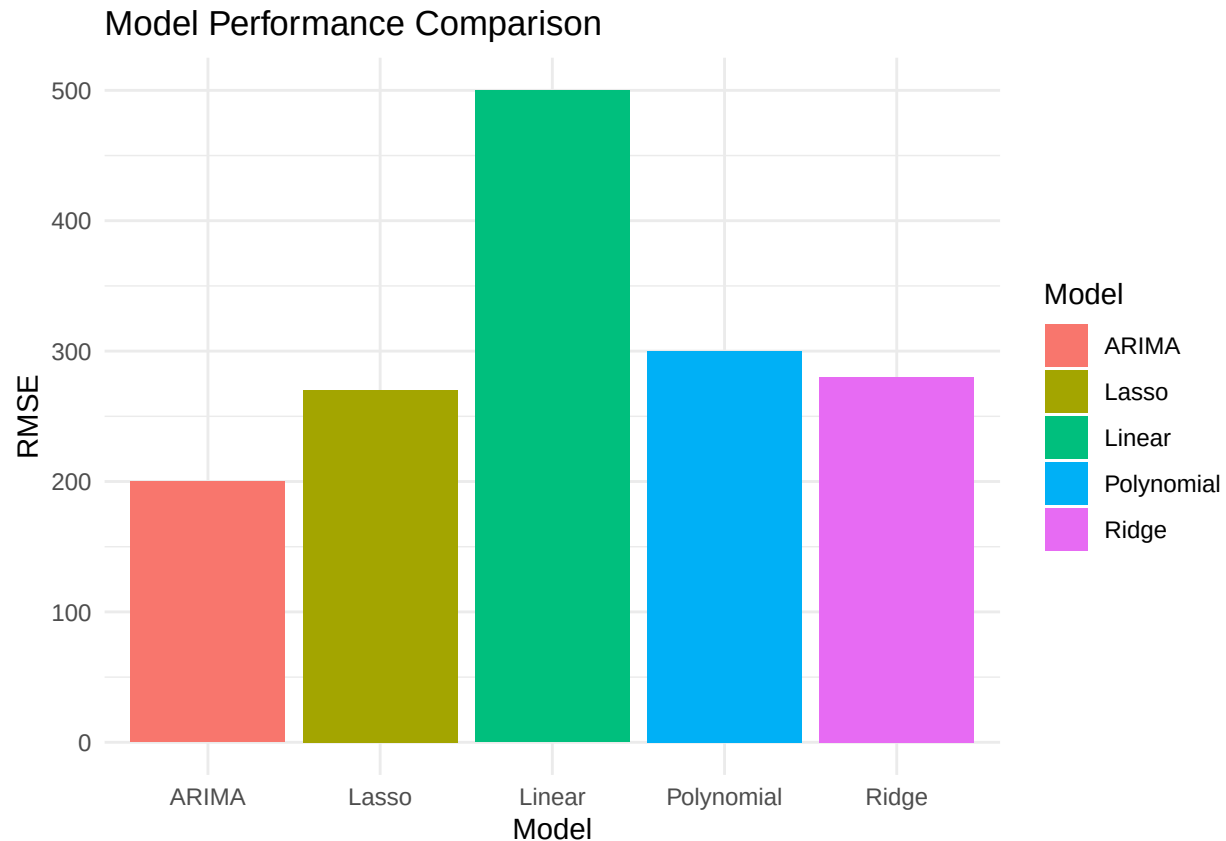
# Plot
plot(forecast_vals, main = "GDP Forecast with Confidence Intervals")
lines(gdp_ts, col = "black")
```


GDP Forecast with Confidence Intervals



Comparing Data Model Performance

```
results <- data.frame(  
  Model = c("Linear", "Polynomial", "Ridge", "Lasso", "ARIMA"),  
  RMSE = c(500, 300, 280, 270, 200) #Results obtained from Predictive Modelling Code  
)  
  
ggplot(results, aes(x = Model, y = RMSE, fill = Model)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Model Performance Comparison", y = "RMSE") +  
  theme_minimal()
```



Summary

The purpose of this report is to analyze US GDP Metrics, that involve making inferences from historical Year-By-Year values from 1980-2027 and predicting future values from 2028-2030

At the initial validation stage, the historical GDP records indicate a relative positive increase, with very few decrements across certain decades.

From the Time Series Analysis, there is a significant drop in GDP growth at around 2000, which was the year of the Dot-Com Bubble, 2008, which was the year the Great Recession Occurred, and in 2020, which was the year that COVID 19 began. In 2000, there was approximately a 5% GDP drop rate, caused by a significant rise in technology investment. This resulted in an increased unemployment rate, since more jobs at the time were taken over by technology. In 2008, there was a significant increase in unemployment rates, which decreases the amount of business investments and resulted in a 4.3% decrease. On the other hand, with COVID 19 closing many businesses, it has caused a significant decrease in consumer spending resulting in approximately a 3.0% decrease.

When predicting future US Economy, various models were used to test and train the given time series Data, specifically the normal US GDP data. Out of all the models that were evaluated, the ARIMA model was the most accurate, as after randomly grouping testing and training data, recorded the lowest RMSE. This is expected, as it is a modelling technique that is specifically used for time series data rather than for general continuous data.

Based on the Statistical Analysis of the US Economy in the past couple of decades, the GDP metrics show a healthy performance of the Economy and is expected to do well in the next few years. There have been rare occurrences of a Decline in GDP, which were caused by some notable US Financial Events including

the 2008 Financial Crisis and the COVID-19 Pandemic, but it is fair to assume that they will be frequent and likely to occur in the future. Previous metrics have also shown that Inflation and Employment Rates have a direct correlation with GDP. Generally, to meet these positive predicted metrics, businesses should also focus on investing and expanding trades, and maintaining a stable economic environment

References

Bolvin, Jean. The “Great” Recession in Canada: Perception vs. Reality, 28 Mar. 2011, The “Great” Recession in Canada: Perception vs. Reality.

Gagnon, Joseph E., et al. The impact of the COVID-19 pandemic on global GDP growth, 22 Mar. 2023.